

MS7200
HDMI to Digital out
Register Map

Note:

- MS7200 registers not in the following list should not be written.
- MS7200 HDMI controller should be accessed through **i2c burst mode**.

Example:

```
VOID mculib_i2c_burstread _16bidx8bval(UINT8 u8_address, UINT16 u16_index, UINT16 u16_length, UINT8 *pu8_value)
{
    UINT16 i;
    _i2c_start();
    if (!_i2c_write_byte(u8_address))
        goto STOP;
    if (!_i2c_write_byte( (UINT8)(u16_index)))
        goto STOP;
    if (!_i2c_write_byte( (UINT8)(u16_index >> 8)))
        goto STOP;
    _i2c_start();
    if (!_i2c_write_byte(u8_address | 0x01))
        goto STOP;
    for (i = 0; i < (u16_length-1); i++)
    {
        *pu8_value++ = _i2c_read_byte(FALSE);
    }
    *pu8_value = _i2c_read_byte(TRUE);
    _i2c_stop();
}
```

Comment

u8_address: chip slave address, either 0x56 or 0xb2:

I2C Read/Write slave address

SA	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0	0	1	0	1	0	1	1	0
1	1	0	1	1	0	0	1	0

u16_index: register address, the value should be 0x2000+4* (INT8) N

u16_length: The value is always 4.

Top MISC					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
0xc2	i2s_mclk_sel	[7..6]	i2s mclk output divide: 00:awalys 1 01:divide 2 10:divide 4 11:divide 8	R/W	0x0
	i2s_mclk_inv	[5]	1:i2s mclk invert	R/W	0x0
	i2s_mclk_en	[4]	1:i2s mclk enable	R/W	0x0
	audio_out_en	[2]	1:RX audio clock enable	R/W	0x0
0xa4	3dr_pad	[4]	1:3dr signal to i2sd1	R/W	0x0
0xa5	mclk_sclk	[2]	1: mclk_sclk I/O use to mclk 0: mclk_sclk I/O use to sclk	R/W	0x0
	spdif_ws	[0]	1:ws_spdif I/O output spdif 0:ws_spdif I/O output i2s_ws	R/W	0x0
0xa8	pclk_drv	[1..0]	3.3V Power: 00->11:6mA,12mA,18mA,24mA 2.5V Power 00->11:4mA,9mA,13mA,18mA 1.8V Power 00->11:3mA,6mA,9mA,12mA	R/W	0x3
0xb0	pattern_sel	[4..3]	test pattern select: 00:Gray pattern 01:Point interlace pattern 10:Horizontal interlace pattern 11:Vertical interlace pattern	R/W	0x0
	test_pattern_en	[0]	1:test pattern enable	R/W	0x0
0xb1	r_val_en	[6]	1:R channel value program enable	R/W	0x0
	b_val_en	[5]	1:B channel value program enable	R/W	0x0
	g_val_en	[4]	1:G channel value program enable	R/W	0x0
0xb8	g_val	[7..0]	G channel value program	R/W	0x0
0xb9	b_val	[7..0]	B channel value program	R/W	0x0
0xba	r_val	[7..0]	R channel value program	R/W	0x0
0xf0	yg_blank_sel	[5]	Y/G blank data select if blank data cut is enable 0:blank data is 0 for 8 bit 1:blank data is 16 for 8 bit	R/W	0x0
	csc_blank_cut	[4]	1:enable(blank data set to a fixed value) 0:disable	R/W	0x0
	dst_csc	[3..2]	output color space conversion 00:RGB 01:YUV444 10:YUV422	R/W	0x0

Top MISC					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
			11:Not used		
	src_csc	[1..0]	input color space conversion 00:RGB 01:YUV444 10:YUV422 11:Not used	R/W	0x0

DVOUT					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
0x09	dvout_rstb	[1]	1:digital out module reset	R/W	0x0
0x16	dvout_en	[1]	1:d[0..23],HS,VS,DE,PCLK output enable	R/W	0x0
0xc0	dvout_div	[7:4]	dvout video divider for pixel repeat 0000:no divide 0001:divide 2 0010:divide 3 ... 1111:divide 16	R/W	0x0
	dvout_clk_div	[3]	1:8-bit and 16-bit DDR mode clock	R/W	0x0
	dvout_clk_sel	[2]	1:8-bit SDR and DDR mode clock	R/W	0x0
0xc1	dvout_lat_div	[3..0]	dvout video lat clock divider 0000:no divide 0001:divide 2 0010:divide 3 ... 1111:divide 16	R/W	0x0
0xe0	fid_in_flip	[7]	1:DVOUT field signal polarity flip	R/W	0x0
	vs_in_flip	[6]	1:DVOUT VS signal polarity flip	R/W	0x0
	hs_in_flip	[5]	1:DVOUT HS signal polarity flip	R/W	0x0
	dvo_bit_remap2	[4]	1:Digital output bit 23~16 remap	R/W	0x0
	dvo_bit_remap1	[3]	1:Digital output bit 15~18 remap	R/W	0x0
	dvo_bit_remap0	[2]	1:Digital output bit 7~0 remap	R/W	0x0
0xe1	dvo_blk_clip_off	[5]	0:turn on clipping for blanking data, clip to 0 in RGB out, in YUV out, Y clip to 0, UV clip to 0x80 1:disable blanking data clipping	R/W	0x1
	dvo_yuv_out	[4]	0: DVO ouput RGB,blanking data clip to 0 1:DVOUT output is YUV, Yclip to 0, UV clip to 0x80	R/W	0x0

DVOUT					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
0xe2	dvo_bit_remap	[7]	1:digital output bit 23~0 remap	R/W	0x0
	dvo_y_uv_flip	[6]	1:digital output Y_UV flip	R/W	0x0
	dvo_uv_flip	[5]	1:digital output UV flip	R/W	0x0
	dvo_2xembedded_en	[4]	1:digital output 2x embedded SYNC enable	R/W	0x0
	dvo_embedded_en	[3]	1:digital output embedded SYNC enable	R/W	0x0
	dvo_8b_en	[2]	1:digital output 8bit mode enable	R/W	0x0
	dvo_16b_en	[1]	1:digital output 16bit mode enable	R/W	0x0
	dvo_en	[0]	1:digital output enable	R/W	0x0
0xe3	dvo_byte_sel	[5..4]	00:dvo_byte_remap[23:16] = dvo_temp[23:16] 01:dvo_byte_remap[23:16] = dvo_temp[15:8] 10:dvo_byte_remap[23:16] = dvo_temp[7:0]	R/W	0x0
	dvo_byte_sel	[3..2]	00:dvo_byte_remap[15:8] = dvo_temp[23:16] 01:dvo_byte_remap[15:8] = dvo_temp[15:8] 10:dvo_byte_remap[15:8] = dvo_temp[7:0]	R/W	0x1
	dvo_byte_sel	[1..0]	00: dvo_byte_remap[7:0] = dvo_temp[23:16] 01: dvo_byte_remap[7:0] = dvo_temp[15:8] 10: dvo_byte_remap[7:0] = dvo_temp[7:0]	R/W	0x2
0xe4	dvo_yuv444_yuv422	[4]	1: two cycles as one pixel U or V. used for the case that clock rate is double of data 0: one cycle as one pixel U or V. used for the case that clock rate is 1x of data.	R/W	0x0
0xe5	12bit_en	[1]	12bit YUV/RGB 444 ddr mode enable	R/W	0x0
	16bit_bt1004_en	[0]	16bit BTA1004 format enable	R/W	0x0
0x1281	pa_s	[2]	clock invert	R/W	0x0
		[1..0]	phase control bits : 00:bypass 01:1ns 10:2ns 11:3ns	R/W	0x0

HDMI RX MISC					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
0x03	rx_pll_calib_clk_en	[7]	RX PLL DIG calibration clock enable	R/W	0x0
	rx_bus_clk_en	[6]	RX bus clock enable	R/W	0x0
	rx_pixel_clk_en	[1]	rx pixel clock enable control 1:enable 0:disable	R/W	0x0

HDMI RX MISC					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
	rx_tmnds_clk_en	[0]	rx tmnds clock enable control 1: enable 0:disable	R/W	0x0
0x04	rx_tmnds_clk_sel	[3..2]	RX tmnds clock select 00: tmnds clock from channel 0 PLL tmnds clock 01: tmnds clock from channel 1 PLL tmnds clock 10: tmnds clock from channel 2 PLL tmnds clock 11: tmnds clock from tmnds link bus clock	R/W	0x0
	rx_audio_clk_en	[0]	RX audio clock enable 0:reset	R/W	0x0
0x09	audll_sw_rstb	[7]	audio DLL clock generation logic reset 0:reset	R/W	0x0
	aupll_ctrl_sw_rstb	[6]	audio PLL control logic reset 0:reset	R/W	0x0
	phy_pllcali_sw_rstb	[5]	PI RX PLL calibration logic reset 0:reset	R/W	0x0
	phy_rx_rcv_sw_rstb	[2]	PI RX receiver logic reset 0:reset	R/W	0x0
	dvout_sw_rstb	[1]	Digital video output module reset 0:reset	R/W	0x0
0x0a	rx_edid_sw_rstb	[2]	RX EDID reset 0:reset	R/W	0x0
0x0c	mcu_access_edid	[7]	MCU access edid enable 1:MCU can read/write edid ram 0:DDC can read edid ram(if edid software reset is released)	R/W	0x01
	rx_audll_div4_en	[6]	RX audio dll output clock divide 4 enable	R/W	0x0
	rx_audll_div2_en	[5]	RX audio dll output clock divide 2 enable(64*fs)	R/W	0x0
	rx_audll_4xen	[4]	RX audio dll output clock 4X enable	R/W	0x0
	rx_aupll_sel[1:0]	[3..2]	RX audio pll select : 00: use AUDIO PLL 01: use AUDIO DLL + AUDIO PLL 10: use AUDIO DLL 11: use AUDIO PLL(XTAL input)	R/W	0x0
	rx_audio_clk_div2_en	[1]	1:rx_audio_clk divide 2 enable	R/W	0x0

HDMI RX MISC					
BASE Address: 0x0000					
Offset	Register name	BIT	Description	Type	Default value
0x15	5v_det_deb_ctrl	[7..4]	RX 5V input debounce control : 0: no debounce else debounced glitch width is $2^{(5v_det_deb_ctrl[3:0])}$ bclk cycles	R/W	0x0
	5v_plug_intr	[3]	RX 5V connect interrupt status(from unplug to plug)	RO	0x0
	5v_unplug_intr	[2]	RX 5V connect interrupt status(from plug to unplug)	RO	0x0
	5v_det_intr_clr	[1]	RX 5V connect interrupt clear, write 1 to clear the interrupt status	WO	0x0
	5v_det	[0]	RX detect 5V power from HDMI TX or not 1: HDMI TX 5V power is connect 0: HDMI TX 5V power is not connect	RO	0x0
0x18	into_en	[5]	interrupt to pad enable, 0: output enable 1: disable, output signal is always 1(before invert)	R/W	0x0
	into_inv	[4]	interrupt to pad_int_out invert, 0: interrupt active low 1: interrupt active high	R/W	0x0
0x36	rx_hs_width	[7..0]	rx_hs_width[7..0]	RO	0x0
0x37	rx_hs_width	[7..0]	rx_hs_width[15..8]	RO	0x0
0x3e	rx_vs_width	[7..0]	rx_vs_width[7..0]	RO	0x0
0x3f	rx_vs_width	[7..0]	rx_vs_width[15..8]	RO	0x0
0x40	rx_vback	[7..0]	rx_vback_width[7..0]	RO	/
0x41	rx_vback	[7..0]	rx_vback_width[15..8]	RO	/
0x50	rx_aud_ch_status	[7..0]	audio channel status bit[7..0]	RO	/
0x52	rx_aud_ch_status	[7..0]	audio channel status bit[23..16]	RO	/
0x53	rx_aud_ch_status	[7..0]	audio channel status bit[31..24]	RO	/
0x54	rx_aud_ch_status	[7..0]	audio channel status bit[39..32]	RO	/
0x77	rx_audio_drv	[3..2]	IO output drive strength select for audio pads: 00: 2mA 01: 4mA 10: 6mA 11: 8mA	R/W	0x1

HDMI RX RCV					
BASE Address: 0x0200					
Offset	Register name	BIT	Description	Type	Default value
0x00	pi_rx_cdr_rstn	[2..0]	RX receiver CDR reset for tmds channel 0/1/2 0:CDR reset 1:CDR normal work	R/W	0x0
0x04	pi_dcmflt_bypss	[4]	1:CDR decimation bypass	R/W	0x0
	pi_dcm_sel	[1..0]	CDR decimation filter select 00:no decimation 01:half decimation 10:1/4 decimation 11:no decimation	R/W	0x0
	pi_det_th	[2..0]	PI phase detection threshold	R/W	0x4
0x15	rx_fifo_rst	[0]	tmds data fifo reset 1: reset fifo 0: fifo normal work	R/W	0x0

HDMI RX PHY					
BASE Address: 0x1000					
Offset	Register name	BIT	Description	Type	Default value
0x00	rxpll_ichp_sel	[7..4]	RXPLL charge pump current select: 4'b0000: 0 4'b0001: 2.5u 4'b0010: 5u 4'b1111: 37.5u	R/W	0x2
	rxpll_rst_vco	[3]	RXPLL VCO reset: 0: normal work 1: reset	R/W	0x0
	rxpll_vreg_pd	[2]	RXPLL ldo power down 0: normal work 1: power down	R/W	0x1
	rxpll_bias_pd	[1]	RXPLL bias power down 0: normal work 1: power down	R/W	0x1
	rxpll_pd	[0]	RXPLL power down 0: normal work 1: power down	R/W	0x1

HDMI RX PHY					
BASE Address: 0x1000					
Offset	Register name	BIT	Description	Type	Default value
0x03	rxpll_bp	[6..4]	RXPLL filter cap select bit0: 125f bit1: 250f bit2: 250f	R/W	0x6
	rxpll_br	[3..2]	RXPLL filter cap select 2'b00: 90Kohm 2'b01: 60Kohm 2'b1x: 30Kohm	R/W	0x1
	rxpll_prediv	[1..0]	RXPLL pre-division 2'b00: /1 2'b01: /2 2'b10: /4 2'b11: /8	R/W	0x0
0x04	rxpll_freq_set	[1..0]	RXPLL output clock divider 2'b00: when TMDS clock > 150MHz 2'b01: when 50MHz ≤ TMDS clock ≤ 150MHz 2'b11: when TMDS clock < 50MHz	R/W	0x1
0x05	rx_pll_kbank_div2_en	[2]	Calibration div2 function enable 0:disable 1:enable	R/W	0x1
0x0f	rxpll_lut_sw_set_en	[2]	RXPLL look up table select, =1: set by software =0: set by hardware build in look up table related register: RG_PLLx_BR[1:0], RG_PLLx_BP[2:0], RG_PLLx_ICHP_SEL[3:0], RG_PLLx_FREQ_SET[3:0], RG_PLLx_PRE_DIV[1:0],	R/W	0x1
0x10	rxpll_cal_trig	[0]	RXPLL auto calibration trigger in auto calibration mode, write 1 to this register will trigger RXPLL power on and calibration flow	W1T	/
0x16	rxpll_lock	[1]	RXPLL lock status 1:lock 0:unlock	RO	/
	rxpll_cal_done	[0]	RXPLL auto calibration done flag 1:done 0:not ready	RO	/
0x18	cal_ref_freq	[5..0]	PLL calibration reference clock frequency	RO	/

HDMI RX PHY					
BASE Address: 0x1000					
Offset	Register name	BIT	Description	Type	Default value
			result bit[13..8]		
0x20	rg_pi_mix_res	[5..4]	PI mixer termination resistor control	R/W	0x0
	rg_pi_mix_cap	[2..0]	PI mixer loading control	R/W	0x0
0x21	rg_pi_mix_isel	[3..0]	PI mixer current control	R/W	0x0
0x22	rg_pi_mux_res	[5..4]	PI mux termination resistor control	R/W	0x0
	rg_pi_mux_cap	[2..0]	PI mux loading control	R/W	0x0
0x23	rg_pi_mux_isel	[3..0]	PI mux current control	R/W	0x0
0x24	rg_pi_rst_n	[6..4]	PI reset 000: reset 111: normal work	R/W	0x0
	rg_pi_pwd	[2..0]	PI power down 000: reset 111: power down	R/W	0x7
0x26	rg_rx_ofs_dec_ch0	[7..4]	CH0 offset decrease control, 10mv/step	R/W	0x0
	rg_rx_ofs_inc_ch0	[3..0]	CH0 offset increase control, 10mv/step	R/W	0x0
0x27	rg_rx_ofs_dec_ch1	[7..4]	CH1 offset decrease control, 10mv/step	R/W	0x0
	rg_rx_ofs_inc_ch1	[3..0]	CH1 offset increase control, 10mv/step	R/W	0x0
0x28	rg_rx_ofs_dec_ch2	[7..4]	CH2 offset decrease control, 10mv/step	R/W	0x0
	rg_rx_ofs_inc_ch2	[3..0]	CH2 offset increase control, 10mv/step	R/W	0x0
0x29	rg_rx_ofs_dec_ck	[7..4]	Clock offset decrease control, 10mv/step	R/W	0x0
	rg_rx_ofs_inc_ck	[3..0]	Clock offset increase control, 10mv/step	R/W	0x0
0x2a	rg_rx_ofs_cal_en_ch2	[2]	CH2 offset calibration enable 0: disable 1: enable	R/W	0x0
	rg_rx_ofs_cal_en_ch1	[1]	CH1 offset calibration enable 0: disable 1: enable	R/W	0x0
	rg_rx_ofs_cal_en_ch0	[0]	CH0 offset calibration enable 0: disable 1: enable	R/W	0x0
0x2b	rg_rx_eq_ch1	[6..4]	CH1 EQ control 3'b000: minimum EQ gain 3'b111: maximum EQ gain	R/W	0x4
	rg_rx_eq_ch0	[2..0]	CH0 EQ control 3'b000: minimum EQ gain 3'b111: maximum EQ gain	R/W	0x4

HDMI RX PHY					
BASE Address: 0x1000					
Offset	Register name	BIT	Description	Type	Default value
0x2c	rg_rx_gain_buf	[5]	Buffer gain control 0: low gain 1: high gain	R/W	0x0
	rg_rx_gain_boost	[4]	EQ DC gain control 0: datarate> 1.5Gbps 1: datarate<=1.5Gbps	R/W	0x0
	rg_rx_eq_ch2	[2..0]	CH0 EQ control 000: minimum EQ gain 111: maximum EQ gain	R/W	0x4
0x2d	rg_rx_vga	[7..6]	VGA gain control 00: minimum 11: maximum	R/W	0x2
	rg_rx_dc_byp_en	[5]	DC test bypass 0: not bypass 1: bypass	R/W	0x0
	rg_rx_rst_n_dc_div	[4]	deep color divider reset 0: reset 1: not reset	R/W	0x0
	rg_rx_ctle_cap_en	[3..2]	EQ cap control 00: ≥2.25Gbps 01: 2.25Gbps > DR≥1.5Gbps 10: 1.5Gbps > DR≥750Mbps 11: DR < 750Mbps	R/W	0x0
	rg_rx_rst_n_div	[1]	DIV5 reset 0: reset 1: normal work	R/W	0x1
	rg_rx_en	[0]	RX enable 0: power down 1: normal work	R/W	0x0
0x2e	rx_ovl_port_dis	[0]	RX termination protect disable 0: protect if single ended voltage [VCC-1.2V]. 1: not protect	R/W	0x0
0x30	rxcn_ovl	[7]	Clock channel negative input overload	RO	/
	rxcp_ovl	[6]	Clock channel positive input overload	RO	/
	rx2n_ovl	[5]	Data channel negative input overload	RO	/
	rx2p_ovl	[4]	Data channel positive input overload	RO	/
	rx1n_ovl	[3]	Data channel negative input overload	RO	/
	rx1p_ovl	[2]	Data channel positive input overload	RO	/
	rx0n_ovl	[1]	Data channel negative input overload	RO	/

HDMI RX PHY					
BASE Address: 0x1000					
Offset	Register name	BIT	Description	Type	Default value
	rx0p_ovl	[0]	Data channel positive input overload	RO	/
0x3b	rg_rxpll_4xdiv_rstn	[1]	RXPLL 4x clock divider reset 0:reset 1:normal work	R/W	0x0

HDMI RX Controller					
BASE Address: 0x2000					
Offset	Register name	BIT	Description	Type	Default value
0x000	hot_plug_detect	[0]	0: receiver not ready 1: receiver ready	R/W	0x0
0x090	vs_pol_adj_mode	[4..3]	VS polarity adjustment 0: Manual. Inverted VSYNC 01: Manual, No VSYNC inversion 10: Automatic 11: Same as 1, Manual, No VSYNC inversion	R/W	0x2
	hs_pol_adj_mode	[2..1]	HS polarity adjustment 00: Manual. Inverted HSYNC 01: Manual, No HSYNC inversion 10: Automatic 11: Same as 1, Manual, No HSYNC inversion	R/W	0x2
0x0bc	dcm_current_mode_sts	[7..4]	Current deep color mode depth indicator 0000: Deep color mode not indicated 0001-0011: reserved 0100: 24 bit 0101: 30 bit 0110: 36 bit 0111: 48 bit 1000-1111: reserved	RO	/
0x0bd	vs_pol_adj_sts	[1]	VS polarity adjust status 0: not active 1: active	RO	/
	hs_pol_adj_sts	[0]	HS polarity adjust status 0: not active 1: active	RO	/
0x148	hs_clk	[7..0]	Total time in horizontal imodetclk clock accumulated over 32 lines	RO	/

HDMI RX Controller					
BASE Address: 0x2000					
Offset	Register name	BIT	Description	Type	Default value
0x149	hs_clk	[15..8]	Total time in horizontal imodetclk clock accumulated over 32 lines 16*xtal_freq/md_ht(u32_value)	RO	/
0x14c	hofs_pix	[7..0]	Horizontal offset time in pixel clocks	RO	/
0x14d	hofs_pix	[15..8]	Horizontal offset time in pixel clocks	RO	/
0x14e	htot_pix	[7..0]	Total horizontal sync time in pixel clocks	RO	/
0x14f	htot_pix	[15..8]	Total horizontal sync time in pixel clocks	RO	/
0x150	hact_pix	[7..0]	horizontal active line	RO	/
0x151	hact_pix	[15..8]	horizontal active line	RO	/
0x168	vact_lin	[7..0]	Vertical active line	RO	/
0x169	vact_lin	[15..8]	Vertical active line	RO	/
0x170	vtot_lin	[7..0]	Vertical total line	RO	/
0x171	vtot_lin	[15..8]	Vertical total line	RO	/
0x180	ilace_sts	[3]	Video interlace or progressive active status 0: progressive 1: interlace	RO	/
	de_act_sts	[2]	Active DE status 0: inactive 1: active	RO	/
	vs_act_sts	[1]	Active VSYNC status 0: inactive 1: active	RO	/
	hs_act_sts	[0]	Active HSYNC status 0: inactive 1: active	RO	/
0x181	rx_stb_sts	[6]	rx timing stable status: 1: vs width is stable 0: vs width is unstable	RO	/
		[5]	rx timing stable status: 1: vblank is stable 0: vblank is unstable	RO	/
		[4]	rx timing stable status: 1: vactive is stable 0: vactive is unstable	RO	/
		[3]	rx timing stable status: 1: hfront is stable 0: hfront is unstable	RO	/
		[2]	rx timing stable status: 1: hs width is stable 0: hs width is unstable	RO	/

HDMI RX Controller					
BASE Address: 0x2000					
Offset	Register name	BIT	Description	Type	Default value
		[1]	rx timing stable status: 1: hblank is stable 0: hblank is unstable	RO	/
		[0]	rx timing stable status: 1: hactive is stable 0: hactive is unstable	RO	/
0x240	afif_init	[0]	Clear audio FIFO by resetting the write and read pointer 0: no clear 1: clear	R/W	0x0
0x258	aud_mute_sel	[6..5]	Manual/automatic audio mute control 0: force unmute (overrides all except when underflow condition occurs) 1: automatic mute when exceeding, FIFO thresholds afif_th_max/min 2: automatic mute enable for FIFO over-underflow 3: force mute (overrides all)	R/W	0x2
0x260	i2s_disable	[7..5]	Disable i2s[2:0] outputs 0: audio enabled 1: audio disabled Note: Shadowing is synchronized, as defined in aport_shdw_ctrl.	R/W	0x0
	spdif_disable	[4..1]	Disable spdif[3:0] outputs bit = 0, audio enabled bit = 1, audio disabled Note: Shadowing is synchronized, as defined in aport_shdw_ctrl.	R/W	0x0
	i2s_32_16	[0]	Select 32 or 16 bit I2S output 0: 16 bit (I2S clock frequency is 32fs) 1: 32 bit (I2S clock frequency is doubled: 64fs) Select I2S clock frequency if hbr_mode is enabled and receiving HBR or NL-PCM audio 0: 32 bit, I2S clock frequency is 128fs 1: 32 bit, I2S clock frequency is 256fs	R/W	0x1
0x261	3d_en	[7]	0:3dr output disable 1:3dr output enable active when 3d mode enable	R/W	0x0

HDMI RX Controller					
BASE Address: 0x2000					
Offset	Register name	BIT	Description	Type	Default value
	3d_2d_mode	[6]	0:3D timing output (active when 3D mode enable) 1:3D to 2D timing output (active when 3D mode enable)	R/W	0x0
	i2s_mode	[5..4]	00 and 11: left justified, delay 0T 01 : right justified, delay 0T 10 : i2s mode (has a 1T delay)	R/W	0x2
0x368	vsi_payload_0	[7..0]	Packet VSI Vendor Specific Payload byte 0	RO	/
0x369	vsi_payload_1	[7..0]	Packet VSI Vendor Specific Payload byte 1	RO	/
0x36a	vsi_payload_2	[7..0]	Packet VSI Vendor Specific Payload byte 2	RO	/
0x36b	vsi_payload_3	[7..0]	Packet VSI Vendor Specific Payload byte 3	RO	/
0x380	pktdec_avmute	[1]	Set_AVMUTE, set audio video mute 0: disabled 1: enable	RO	/
0x390	pktdec_acr_cts	[7..0]	CTS_decode[7..0]	RO	/
0x391	pktdec_acr_cts	[7..0]	CTS_decode[15..8]	RO	/
0x394	pktdec_acr_n	[7..0]	N_decode[7..0]	RO	/
0x395	pktdec_acr_n	[7..0]	N_decode[15..8]	RO	/
0x3a3	pix_rep_factor	[3..0]	PR3 PR2 PR1 PR0, pixel repetition factor 0000: no repetition, pixel sent once 0001: pixel sent 2 times 0010: pixel sent 3 times 0011: pixel sent 4 times 0100: pixel sent 5 times 0101: pixel sent 6 times 0110: pixel sent 7 times 0111: pixel sent 8 times 1000: pixel sent 9 times 1001: pixel sent 10 times 1010-1111: reserved	RO	/
0x403	ceavid_rst	[7]	Synchronous clear for CEA video interface, active high	R/W	0x0